

UniDyn--Demo-08.nb

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Abstract: This demonstration notebook performs a quantum optics calculation.

Set the path to the package

Check the Mathematica version number .

```
In[175]:= $VersionNumber
```

```
Out[175]:= 12.3
```

Tell *Mathematica* the path to the directory containing the packages and the /studies directory.

EDIT THE FOLLOWING PATH STRINGS:

```
In[176]:= $UniDynPath =  
          "/Users/jam99/Dropbox/MarohnGroup__Software_Library/UniDyn/  
          unidyn";
```

YOU SHOULD NOT NEED TO EDIT ANYTHING FROM HERE ONWARDS.

Load and test the packages

Load Unidyn

Append the package path to the system path. Before trying to load the package, ask *Mathematica* to find it. This is a test that we directed *Mathematica* to the correct directory. The output of this command should be the full system path to the UniDyn.m file.

```
In[177]:= $Path = AppendTo[$Path, $UniDynPath];  
          FindFile["UniDyn`"]
```

```
Out[178]:= /Users/jam99/Dropbox/MarohnGroup__Software_Library/UniDyn/unidyn/UniDyn.m
```

Now that we are confident that the path is set correctly, load the **Unidyn** package. Setting the global \$VerboseLoad variable to True will print out the help strings for key commands in the package.

```
In[179]:= $VerboseLoad = True;
Needs["UniDyn`"]
```

Run the Unidyn unit tests

Carry out the unit tests for the **Unidyn** functions.

```
In[181]:= SetDirectory[$UniDynPath];
In[182]:= fn = FileNames["*-tests.m"];
test$report = TestReport /@ fn;
TableForm[Table[test$report[[k]], {k, 1, Length[test$report]}]]
```

Out[184]//TableForm=

TestReportObject [			Title: Test Report: Comm-tests.m Success rate: 100% Tests run: 23	]
TestReportObject [			Title: Test Report: Evolver1-tests.m Success rate: 100% Tests run: 14	]
TestReportObject [			Title: Test Report: Evolver2-tests.m Success rate: 100% Tests run: 13	]
TestReportObject [			Title: Test Report: Evolve-tests.m Success rate: 100% Tests run: 10	]
TestReportObject [			Title: Test Report: Inv-tests.m Success rate: 96% Tests run: 24	]
TestReportObject [			Title: Test Report: Mult-tests.m Success rate: 100% Tests run: 18	]
TestReportObject [			Title: Test Report: OpQ-tests.m Success rate: 100% Tests run: 21	]
TestReportObject [			Title: Test Report: Osc-tests.m Success rate: 100% Tests run: 22	]
TestReportObject [			Title: Test Report: SpinBoson-tests.m Success rate: 100% Tests run: 13	]
TestReportObject [			Title: Test Report: Spins-tests.m Success rate: 100% Tests run: 14	]

Evolver settings

Choose which **Evolver** to use.

```
In[185]:= Evolver = Evolver2;
```

For the expressions below to simplify, we need another simplification rule for the inverse operator.

```
In[186]:= Inv /: HoldPattern[Inv[Inv[a_?OperatorQ]]] := a
```

Two two-level systems coupled to a common cavity

Clear all the symbols; start afresh.

```
In[187]:= Clear[Ix, Iy, Iz, I$P, I$m, Sx, Sy, Sz, S$p, S$m, aR, aL];
Clear[ω, ω$I, ω$S, g$I, g$S, Δ];
Clear[H$0, H$1];
```

Create pseudo-spin operators for an *I* atom and an *S* atom.

Create raising and lowering operators for the single-mode cavity.

```
In[190]:= TwoSpinBoson$CreateOperators[Ix, Iy, Iz, I$P, I$m, Sx, Sy, Sz, S$p, S$m, aR, aL];
```

... SpinSingle\$CreateOperators: Spin operators already exist.

... SpinSingle\$CreateOperators: Adding spin commutations relations.

... SpinSingle\$CreateOperators: Angular momentum $L = 1/2$. Adding operator simplification rules.

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... SpinSingle\$CreateOperators: Adding spin commutations relations.

... SpinSingle\$CreateOperators: Angular momentum $L = 1/2$. Adding operator simplification rules.

... OscSingle\$CreateOperators: Oscillator operators already exist.

... OscSingle\$CreateOperators: Adding oscillator commutations relations.

... TwoSpinBoson\$CreateOperators: Creating operators.

... TwoSpinBoson\$CreateOperators: Adding I_p , I_m and S_p , S_m commutations relations.

... TwoSpinBoson\$CreateOperators: Adding I_p , I_m and S_p , S_m simplification rules.

... TwoSpinBoson\$CreateOperators: Adding a_L and a_R normal ordering rule.

Scalar parameters appearing the Hamiltonians .

```
In[191]:= CreateScalar[
   $\omega$ , (* cavity frequency *)
   $\omega_I$ , (* atom I frequency *)
   $\omega_S$ , (* atom S frequency *)
   $g_I$ , (* unitless coupling constant characterizing
    the interaction of atom I with the cavity mode *)
   $g_S$ , (* unitless coupling constant characterizing the
    interaction of atom I with the cavity mode *)
   $\Delta$  (* resonance offset *)
]
```

Hamiltonian for the bare atoms / and S and the cavity.

```
In[192]:= H$0 =  $\omega_I$  Iz +  $\omega_S$  Sz +  $\omega$  Mult[aR, aL];
```

Coupling Hamiltonian describing the exchange of photons between atom / and the cavity and atom S and the cavity.

```
In[193]:= H$1 =  $g_I \omega$  (Mult[I$p, aL] + Mult[I$m, aR]) +  $g_S \omega$  (Mult[S$p, aL] + Mult[S$m, aR]);
```

Version 1

The cavity frequency is ω , atom / has resonance frequency ω (in tune with the cavity resonance), and atom S has resonance frequency $\omega - \Delta$ (detuned from the cavity resonance by an amount Δ). Here atoms / and S are different species.

Take the coupling Hamiltonian into the interaction representation:

```
In[194]:= Clear[H$1$int];
H$1$int[t_] =
  (Evolve[H$0, t, H$1] /. Evolve -> Evolver2) /. { $\omega_I \rightarrow \omega$ ,  $\omega_S \rightarrow \omega - \Delta$ } //
  Simplify // Expand
```

```
Out[195]=  $g_I \omega$  Mult[I$m, aR] +  $g_I \omega$  Mult[I$p, aL] +
   $e^{-i t \Delta} g_S \omega$  Mult[S$m, aR] +  $e^{i t \Delta} g_S \omega$  Mult[S$p, aL]
```

Using the Magnus expansion, compute a zeroth-order average Hamiltonian:

```
In[196]:= H$1$int$0 =  $\frac{1}{t_c} \text{Integrate}[H$1$int[t], \{t, 0, t_c\}] /. t_c \rightarrow \frac{2 \pi}{\Delta} // \text{Expand}$ 
```

```
Out[196]=  $g_I \omega$  Mult[I$m, aR] +  $g_I \omega$  Mult[I$p, aL]
```

To zeroth order, only atom / is coupled to the cavity.

Using the Magnus expansion, compute a first-order average Hamiltonian:

$$\text{In[197]:= } H_{1\text{int}} = -\frac{I}{2 t_{\text{sc}}} \text{Integrate}\left[\text{Integrate}\left[\text{Comm}[H_{1\text{int}}[t_1], H_{1\text{int}}[t_2]], \{t_1, 0, t_2\}\right], \{t_2, 0, t_{\text{sc}}\}\right] /. t_{\text{sc}} \rightarrow \frac{2\pi}{\Delta}$$

$$\begin{aligned} \text{Out[197]= } & -\frac{1}{8\pi\Delta} i g_S \omega^2 \left(-4 i g_S \pi - 8 i g_S \pi S_z + 4 i g_I \pi \text{Mult}[S_m, I_p] + 4 i g_I \pi \text{Mult}[S_p, I_m] - \right. \\ & 16 i g_S \pi \text{Mult}[S_z, a_R, a_L] + 2 g_I \text{Mult}[I_m, S_p] \left(1 - \cos\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] + \right. \\ & \left. \frac{2\pi \left(i \cos\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \text{Sign}[\Delta] - \sin\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \right)}{\sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]} - \right. \\ & \left. i \text{Sign}[\Delta] \sin\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \right) + \frac{1}{\text{Abs}[\Delta]} \\ & 2 g_I \text{Mult}[I_p, S_m] \left(\frac{2\pi \left(i \cos\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \text{Sign}[\Delta] + \sin\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \right)}{\sqrt{\frac{1}{\Delta^2}}} + \right. \\ & \left. \left. \text{Abs}[\Delta] \left(-1 + \cos\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] - i \text{Sign}[\Delta] \sin\left[2\pi \sqrt{\frac{1}{\Delta^2}} \text{Abs}[\Delta]\right] \right) \right) \right) \end{aligned}$$

Simplify the first-order average Hamiltonian by assuming that the resonance offset is positive.

Use sorted multiplication to put the I and S angular momentum operators in canonical order, with the I -atom operators first and the S -atom operators second.

$$\text{In[198]:= } \text{ans1} = \text{Expand}[\text{Simplify}[H_{1\text{int}}[1, \text{Assumptions} \rightarrow \{\text{Element}[\Delta, \text{Reals}], \Delta > 0\}]] /. \{ \text{Mult} \rightarrow \text{SortedMult} \}$$

$$\begin{aligned} \text{Out[198]= } & -\frac{g_S^2 \omega^2}{2\Delta} - \frac{g_S^2 S_z \omega^2}{\Delta} + \frac{g_I g_S \omega^2 \text{Mult}[I_m, S_p]}{\Delta} + \\ & \frac{g_I g_S \omega^2 \text{Mult}[I_p, S_m]}{\Delta} - \frac{2 g_S^2 \omega^2 \text{Mult}[S_z, a_R, a_L]}{\Delta} \end{aligned}$$

```
In[199]:= Collect[ans1,  $\frac{\omega^2}{\Delta}$ ]
```

```
Out[199]=  $\frac{1}{\Delta} \omega^2 \left( -\frac{g^2 S^2}{2} - g^2 S^2 Sz + g I g S \text{Mult}[I\$m, S\$p] + \right.$   

 $\left. g I g S \text{Mult}[I\$p, S\$m] - 2 g^2 S^2 \text{Mult}[Sz, aR, aL] \right)$ 
```

There are two physical effects contributed by the first-order terms.

1) A shift in the atom S energy levels, proportional to $(n + 1/2)$, with n the number of photons in the cavity.

2) A coupling between atoms I and S of the form $I_- S_+ + I_+ S_-$.

The coherence I_+ will evolve under this Hamiltonian into $I_+ \cos[\Omega t] + 2 i I_z S_+ \sin[\Omega t]$, entangling the electronic states of atoms I and S .

```
In[200]:= Evolve[ $\Omega$  (Mult[I$m, S$p] + Mult[I$p, S$m]), t, I$p] /. Evolve -> Evolver2
```

```
Out[200]= I$p Cos[t  $\Omega$ ] + 2 i Mult[Iz, S$p] Sin[t  $\Omega$ ]
```

Version 2

The cavity frequency is ω and atom I and S each has a resonance frequency $\omega - \Delta$ (detuned from the cavity resonance by an amount Δ). Here atoms I and S are the same species.

Take the coupling Hamiltonian into the interaction representation:

```
In[201]:= Clear[H$1$int];  
H$1$int[t_] =  
  (Evolve[H$0, t, H$1] /. Evolve -> Evolver2) /. { $\omega$ $I ->  $\omega - \Delta$ ,  $\omega$ $S ->  $\omega - \Delta$ } //  
  Simplify // Expand
```

```
Out[202]=  $e^{-i t \Delta} g I \omega \text{Mult}[I\$m, aR] + e^{i t \Delta} g I \omega \text{Mult}[I\$p, aL] +$   

 $e^{-i t \Delta} g S \omega \text{Mult}[S\$m, aR] + e^{i t \Delta} g S \omega \text{Mult}[S\$p, aL]$ 
```

Using the Magnus expansion, compute a zeroth-order average Hamiltonian:

```
In[203]:= H$1$int$0 = Expand[ $\frac{1}{t\$c}$  Integrate[H$1$int[t], {t, 0, t$c}] /. t$c ->  $\frac{2 \pi}{\Delta}$ ]
```

```
Out[203]= 0
```

Using the Magnus expansion, compute a first-order average Hamiltonian:

$$\text{In[204]:= } H\$1\$int\$1 = -\frac{I}{2 t\$c} \text{Integrate[Integrate[Comm[H\$1\$int[t1], H\$1\$int[t2]], \{t1, 0, t2\}], \{t2, 0, t\$c\}] /. t\$c \rightarrow \frac{2 \pi}{\Delta}}$$

$$\begin{aligned} \text{Out[204]= } & -\frac{1}{8 \pi \Delta} i \omega^2 \left(-4 i \pi \left(g\$I^2 (1 + 2 Iz) + g\$S^2 (1 + 2 Sz) \right) - \right. \\ & 2 g\$I (g\$S (\text{Mult}[I\$p, S\$m] + \text{Mult}[S\$p, I\$m]) + 2 g\$I \text{Mult}[Iz, aR, aL]) - \\ & 4 g\$S^2 \text{Mult}[Sz, aR, aL] + \\ & 2 (g\$I (g\$S (\text{Mult}[I\$m, S\$p] + \text{Mult}[S\$m, I\$p]) + 2 g\$I \text{Mult}[Iz, aR, aL]) + \\ & 2 g\$S^2 \text{Mult}[Sz, aR, aL]) - \\ & 2 i (g\$I (g\$S ((-i + 2 \pi) \text{Mult}[I\$m, S\$p] + i (\text{Mult}[I\$p, S\$m] - \text{Mult}[S\$m, I\$p] + \text{Mult}[\\ & \quad S\$p, I\$m]) + 2 \pi (\text{Mult}[I\$p, S\$m] + \text{Mult}[S\$m, I\$p] + \text{Mult}[S\$p, I\$m])) + \\ & \left. 8 g\$I \pi \text{Mult}[Iz, aR, aL]) + 8 g\$S^2 \pi \text{Mult}[Sz, aR, aL]) \right) \end{aligned}$$

Use sorted multiplication to put the I and S angular momentum operators in canonical order, with the I -atom operators first and the S -atom operators second.

$$\text{In[205]:= } \text{ans2} = (H\$1\$int\$1 // \text{Expand}) /. \{\text{Mult} \rightarrow \text{SortedMult}\}$$

$$\begin{aligned} \text{Out[205]= } & -\frac{g\$I^2 \omega^2}{2 \Delta} - \frac{g\$S^2 \omega^2}{2 \Delta} - \frac{g\$I^2 Iz \omega^2}{\Delta} - \frac{g\$S^2 Sz \omega^2}{\Delta} - \frac{g\$I g\$S \omega^2 \text{Mult}[I\$m, S\$p]}{\Delta} - \\ & \frac{g\$I g\$S \omega^2 \text{Mult}[I\$p, S\$m]}{\Delta} - \frac{2 g\$I^2 \omega^2 \text{Mult}[Iz, aR, aL]}{\Delta} - \frac{2 g\$S^2 \omega^2 \text{Mult}[Sz, aR, aL]}{\Delta} \end{aligned}$$

$$\text{In[206]:= } \text{Collect}[\text{ans2}, \frac{\omega^2}{\Delta}]$$

$$\begin{aligned} \text{Out[206]= } & \frac{1}{\Delta} \omega^2 \left(-\frac{g\$I^2}{2} - \frac{g\$S^2}{2} - g\$I^2 Iz - g\$S^2 Sz - g\$I g\$S \text{Mult}[I\$m, S\$p] - \right. \\ & \left. g\$I g\$S \text{Mult}[I\$p, S\$m] - 2 g\$I^2 \text{Mult}[Iz, aR, aL] - 2 g\$S^2 \text{Mult}[Sz, aR, aL] \right) \end{aligned}$$

There are three physical effects contributed by the first-order terms.

- 1) A shift in the atom S energy levels, proportional to $(n + 1/2)$, with n the number of photons in the cavity.
- 2) A shift in the atom I energy levels, proportional to $(n + 1/2)$, with n the number of photons in the cavity.
- 3) A coupling between atoms I and S of the form $I_- S_+ + I_+ S_-$.